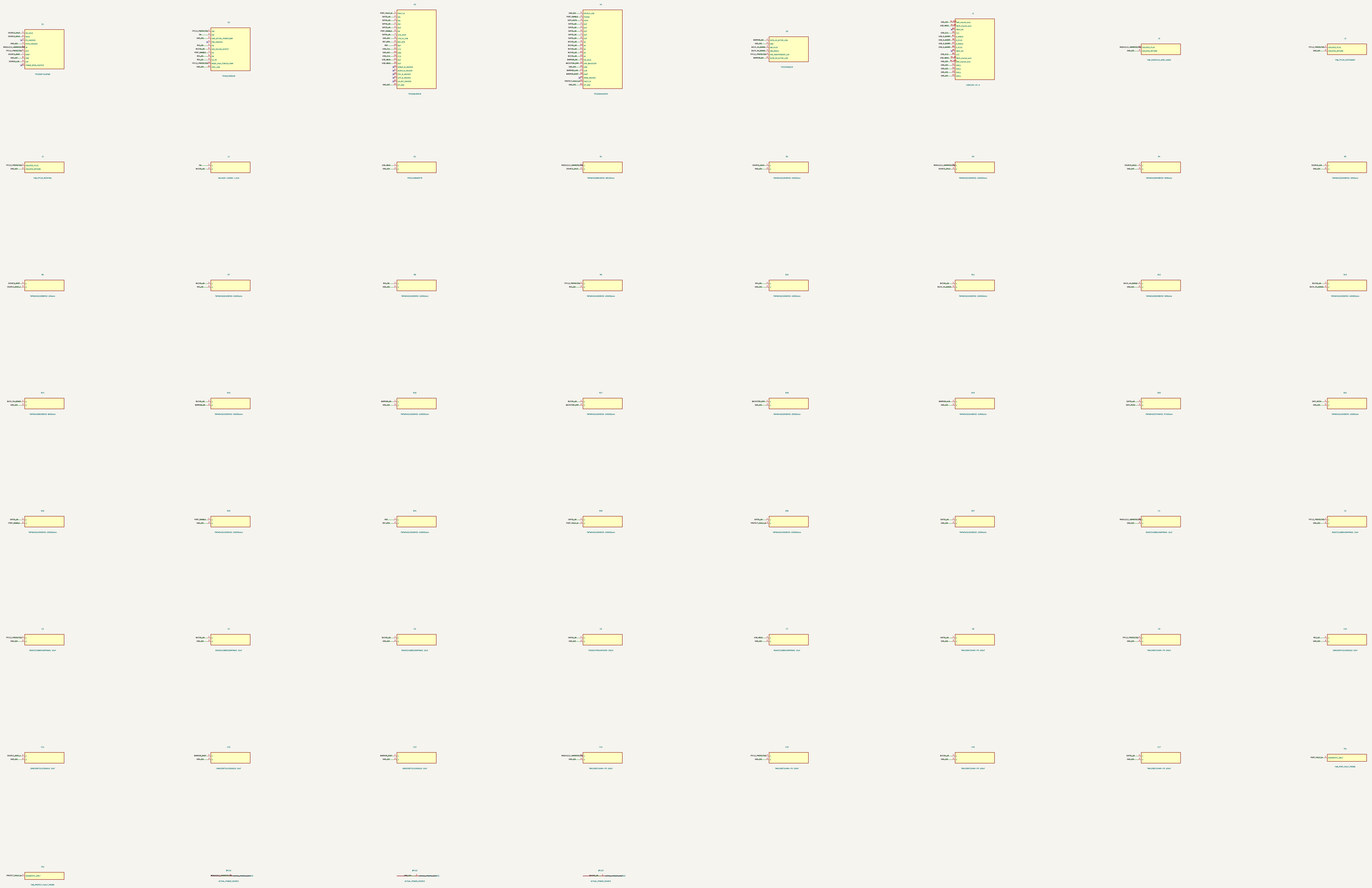

PROTOTYPE / NOT A RELEASE: only isolated12V is shown; no mains wiring. Stationary U1 protects the FPC before the rotary joint.

Rotating U2 TPS62135:4A forcedPWM5.18V. U5 senses unprotected buck voltage; independent U4 disconnects and reverse-blocks before U3.

U5 OUTA+OUTB are wired open-drain AND to U4 EN. No MCU/firmware/PG substitutes for independent voltage sensing.

U4 internal OVP is a higher backstop, not the precision5.4V trip. Response/injected energy and power-ramp extremes remain UNMET.

U3 CHG=SAFE5, CHG_HI=GND:1.5A only. Both CC orientations. Fullcontact GCT USB4105, fourD contacts passive short. No USBdata or higherPD.

FPC ports J2/J3 are actual direct-solder lands on two boards plus an external20mm harness. Their finite resistance is independently budgeted.

Input flag = isolated source atJ0; BUCK flag = actual U2.SW-L1 conversion path; GND flag = external isolated return. No other power flags.

Proposed shell = isolatedGND, never rotatingPE. Mains/module/pod/PE/endcaps are external unqualified integration contracts.

Unrouted placement allocation only. Connector3.46mm max height, clamp/flex reserves and routing/thermal/USB/safety gates remain open.

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Title: NOVA 12V internal bus / independently protected 5V USB prototype

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